

Research Interests

Overview

My research goal is to develop intelligent embodied agents that can perceive, understand, and interact with complex physical environments. I am particularly interested in the intersection of 3D perception, semantic representation learning, world models, and reinforcement learning.

1. Semantic 3D Representation Learning for Embodied Intelligence

Modern robots require robust environmental representations to perform complex tasks in open-world scenarios. I am interested in developing learning-based approaches that combine:

- 3D visual perception
- neural representations
- vision-language models
- semantic scene understanding

to enable robots to acquire structured and generalizable representations of the physical world.

2. World Models for Embodied Agents

I am interested in building predictive world models that can capture both:

- physical dynamics
- high-level semantic information

for improved planning and decision making.

My recent work PrivilegedJEPA explores injecting VLM semantic priors into latent world models, aiming to enhance environment understanding and improve downstream control performance.

3. Learning-based Autonomous Systems

I am interested in combining:

- perception
- representation learning
- reinforcement learning

to develop autonomous agents capable of adapting to complex environments.

Future directions I hope to explore include:

- perception-driven robot learning
- simulation-to-real transfer
- generalizable embodied agents